

Unit V

MOBILE WEB DEVELOPMENT

Mobile-First: What is Mobile-First? What is Mobile Web? Understanding Mobile Devices and Desktop.

JQuery Mobile: Introduction to the jQuery Mobile Framework, Set-up jQuery Mobile, Pages, Icons, Transitions, Layouts Widgets, Events, Forms, Themes, Formatting Lists, Header and Footer, CSS Classes, Data Attributes, Building a Simple Mobile Webpage.

What is Mobile-First Design?

- Mobile-first design is an approach where you design your website for mobile devices first and then for desktops. This method is crucial because most people today use their smartphones and tablets to go online rather than laptops or computers.
- Mobile-first designs ensure that your website not only looks good but also works seamlessly across different devices, providing the best user experience.
- This approach helps increase website traffic and ensures that your site adjusts to different screen sizes effectively.
- To create a successful mobile-first website, consider the following tips:
 - Use large fonts and buttons to make navigation easier on small screens.
 - Keep the text short and to the point to improve readability.
 - Use high-quality images to make the content visually appealing.

Why Mobile-First Design?

- The Mobile-First concept is crucial because it prioritizes the user experience on mobile devices, which are now the most popular way to access the Internet. Designing for mobile-first ensures that your products are optimized for the majority of users, as most internet traffic today comes from smartphones and tablets.
- Mobile devices have small screens and touch interfaces, which can significantly impact the user's experience. By focusing on mobile design first, you ensure that your website or app is intuitive and user-friendly. This approach guarantees that users can easily navigate and interact with your site, no matter the device they are using. In turn, this leads to better user engagement, increased website traffic, and a more successful overall design.

Mobile-First Design Process

The **Mobile-First design process** involves several key steps to ensure your website is optimized for mobile devices first. Here's a simplified and effective approach:

1. Wireframing:

Start by creating wireframes for your layout. Focus on the content and arrange the screen size to suit your needs. This helps in visualizing the structure and flow of your mobile design.

2. Content Inventory:

Compile all the elements you plan to include in a spreadsheet or similar document. This ensures you have a clear overview of all the content.

3. Visual Hierarchy:

Prioritize content items in your inventory to display the most important elements effectively. This helps in guiding users' attention to key information.

4. Design for Small Screens First:

Begin by designing a mobile wireframe. Use this as a model and then expand to larger breakpoints. You can enlarge the screen until there is ample negative space, ensuring a clean and spacious layout.

5. Expand Touch Targets:

Fingers are wider than mouse cursors, so make sure your interactive elements are large enough to tap easily. This improves usability on touch screens.

6. Avoid Relying on Hovers:

Hovers add interactivity, but they don't work on mobile devices. Design interactions that don't rely on hover effects to ensure functionality on all devices.

7. Use Smaller Graphics:

Large graphics like landscape images and complex visuals don't perform well on smaller screens. Opt for smaller, simpler graphics that display well even on mobile devices.

Benefits of Mobile-First Design

Mobile-First Design offers numerous benefits:

1. **Responsive Design:** Ensures your website adapts to any device, from **smartphones** to **desktops**, reaching a wider audience.
2. **Simplified Content:** Streamlines and simplifies information, making it easily accessible on smaller screens.

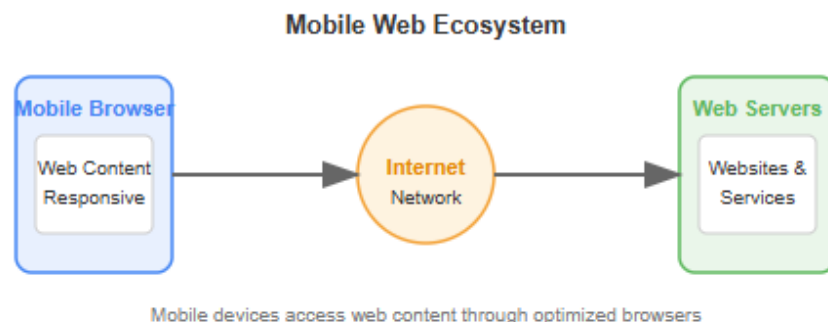
3. **Growing Mobile Usage:** Meets the increasing demand for mobile-friendly websites as more people use their **smartphones** and **tablets** to browse the internet.
4. **Improved SEO:** Enhances your website's SEO ranking, making it more likely to appear in search results and attract more visitors.
5. **Optimized Performance:** Uses smaller, optimized images and efficient code that load quickly on mobile devices, providing a better user experience.

What is Mobile Web?

The Mobile Web is the part of the internet that users access through mobile devices. It includes websites that are:

- Viewable on small screens
- Touch-friendly
- Fast-loading on mobile networks
- Optimized for different screen sizes and resolutions

Instead of downloading an app, users simply open a browser (like Chrome or Safari) and visit a website.



Examples of Mobile Web

- Google search on your phone
- Facebook opened in browser (not app)
- Online shopping websites like Amazon

Key Features of Mobile Web

1. Responsive Design

- Websites automatically adjust layout based on screen size
- Works on phones, tablets, and desktops
- Example: A website rearranges content into a single column on mobile

2. Touch-Friendly Interface

- Buttons are larger and easy to tap
- No need for mouse or keyboard

3. Faster Loading

- Uses optimized images and lightweight code
- Important for slower mobile internet connections

4. No Installation Required

- Users don't need to install anything
- Just open a browser and access instantly

5. Cross-Platform

- Works on Android, iOS, and any device with a browser

6. Location Awareness

- Integration with GPS and location services for context-aware content.

Mobile Web Technologies

Modern mobile web relies on several key technologies:

- HTML5 Provides semantic markup and multimedia support without plugins.
- CSS3 Media Queries Enable responsive design based on screen characteristics.
- Progressive Web Apps (PWA) Bridge the gap between web and native apps with offline capabilities.
- Accelerated Mobile Pages (AMP) Google's framework for lightning-fast mobile page loading.

Advantages of Mobile Web

- Easy access (no install)
- Lower storage usage

- Works on all devices
- Easy to update (changes reflect instantly)

Disadvantages

- May be slower than apps
- Limited access to device features (like camera, GPS – though improving)
- Requires internet connection

Understanding Mobile Devices and Desktop.

Mobile Devices

Mobile devices are portable electronic devices that users can carry anywhere.

Examples

- Smartphones
- Tablets

Key Characteristics

1. Small Screen Size

- Limited display area
- Content must be compact and scrollable

2. Touch Input

- Uses fingers instead of mouse
- Requires large buttons and spacing

3. Portability

- Can be used anywhere (home, travel, office)

4. Limited Resources

- Less processing power than desktops
- Limited battery life

5. Internet Connectivity

- Uses mobile data (4G/5G) or Wi-Fi

Mobile Usage Behavior

- Short sessions
- Quick tasks (search, chat, social media)
- Users prefer fast-loading and simple UI

Understanding Desktop Devices

Desktop devices include computers that are typically used at a fixed location.

Examples

- Desktop computers
- Laptops

Key Characteristics

1. Large Screen Size

- More space for content
- Multiple windows can be used

2. Keyboard & Mouse Input

- Precise control
- Suitable for complex tasks

3. Higher Performance

- More RAM, CPU power
- Handles heavy applications (coding, gaming, design)

4. Stable Power Source

- No battery limitations (especially desktops)

5. Better Multitasking

- Easy to run multiple applications simultaneously

Desktop Usage Behavior

- Longer sessions
- Complex work (coding, editing, analysis)
- Detailed interaction

Mobile Devices vs Desktop

Feature	Mobile Devices	Desktop Devices
Screen Size	Small	Large
Input	Touch	Keyboard + Mouse
Portability	High	Low
Performance	Moderate	High
Battery	Limited	Unlimited (plugged in)
Usage Type	Quick tasks	Complex tasks
Internet	Mobile data + Wi-Fi	Mostly Wi-Fi / LAN

JQuery Mobile

- JQuery Mobile is a user interface framework, which is built on jQuery Core and used for developing responsive websites or applications that are accessible on mobile, tablet, and desktop devices.
- It uses the features of both jQuery and jQuery UI to provide API features for mobile web applications.
- It was developed by the jQuery project team in the year 2010 and written in JavaScript.

Features of jQuery Mobile

- It is built on jQuery Core and "write less, do more" UI framework.
- It is an open source framework, and cross-platform as well as cross-browser compatible.
- It is written in JavaScript and uses features of both jQuery and jQuery UI for building mobile-friendly sites.

- It integrates HTML5, CCS3, jQuery and jQuery UI into one framework for creating pages with minimal scripting.
- It includes Ajax navigation system that uses animated page transitions.

Set-up jQuery Mobile

Setting up jQuery Mobile is simple because it is a front-end framework that works using HTML, CSS, and JavaScript.

There are two ways to set up jQuery Mobile:

Method 1: Using CDN

Add the following links inside your HTML file:

```
<!DOCTYPE html>

<html>

<head>

  <title>jQuery Mobile Setup</title>

  <!-- jQuery Library -->

  <script src="https://code.jquery.com/jquery-1.11.1.min.js"></script>

  <!-- jQuery Mobile CSS -->

  <link rel="stylesheet" href="https://code.jquery.com/mobile/1.4.5/jquery.mobile-1.4.5.min.css">

  <!-- jQuery Mobile JS -->

  <script src="https://code.jquery.com/mobile/1.4.5/jquery.mobile-1.4.5.min.js"></script>

  <!-- Viewport for Mobile -->

  <meta name="viewport" content="width=device-width, initial-scale=1">

</head>

</html>
```


Method 2: Download and Use Locally

1. Download files from:
 - jQuery
 - jQuery Mobile
2. Save them in your project folder:

```
project/
├── index.html
├── css/
│   └── jquery.mobile.css
├── js/
│   ├── jquery.js
│   └── jquery.mobile.js
```

3. Link them in HTML:

```
<link rel="stylesheet" href="css/jquery.mobile.css">
<script src="js/jquery.js"></script>
<script src="js/jquery.mobile.js"></script>
```

jQuery Mobile - Pages

- The user can interact with jQuery Mobile pages, which groups the content into logical views and page views.
- Page view can be animated using page transitions.
- Multiple pages can be created using HTML document and therefore, there is no need of requesting the content from the server.

Types of pages

1. **Single Page**- A single page is created in HTML document using a standard way of writing a template.
2. **Multi-Page Template** -Multiple pages can be included in the single HTML document, which loads together by adding multiple divs with data-role = "page".
3. **Dialogs Page** -Modal dialogs open content in an interactive overlay above the page.

Example

```
<div data-role="page" id="home">
```

```
<div data-role="header"><h1>Header</h1></div>
<div data-role="content"><p>Content</p></div>
<div data-role="footer"><h4>Footer</h4></div>
</div>
```

jQuery Mobile - Icons

jQuery Mobile provides a set of built-in icons, which can be used with buttons, listview buttons which will make the buttons more attractive.

type of Icons

- 1. Icon Set** - It sets the icon in the button.
- 2 Positioning Icons** - It specifies the position of the icon in the button.
- 3 Icon-only** - It displays only an icon in the button.
- 4 Icon shadow** - It adds an icon shadow in your button.
- 5 Removing Circle** - It removes the grey circle around the icon.
- 6 Black or White Icons** - It changes the color of icons to black or white.
- 7 Combining alt and nodisc** - It combines the alt and nodisc classes to the icon.

Example

```
<a href="#" data-role="button" data-icon="home">Home</a>
```

jQuery Mobile - Transitions

It allows to change property values which occurs over specified duration and alters behavior of an element from one state to another state by applying different styles for each state.

Following some of the page transitions used in the jQuery Mobile framework –

- 1 fade** -You can make elements fade in and out of visibility.
- 2 flip** -Flip the elements from back to front to the next page.
- 3 pop** -You can create a popup window.
- 4 flow** -Display the next page by keeping current page away.
- 5 slide** -You can slide the page from right to left.

6 slidefade -Slides the page from right to left and fades in the next page.

7 slideup -Slides the page from bottom to up.

8 slidedown -Slides the page from top to bottom.

9 turn -You can turn to the next page.

10 none -You cannot use any transition effect by using this attribute.

Example

```
<a href="#page2" data-transition="slide">Go</a>
```

jQuery Mobile - Layouts

Grid systems are used to create page layouts through a series of rows and columns that house your content.

1 Grid -jQuery mobile grid system creates page layouts through a series of rows and columns.

2 Buttons in grids - Collection of button in grid format in jQuery mobile.

3 Custom responsive grid - The basic grid style can be easily extended to the custom responsive layout using media queries in CSS.

Example

```
<div class="ui-grid-a">  
  <div class="ui-block-a">Block A</div>  
  <div class="ui-block-b">Block B</div>  
</div>
```

jQuery Mobile - Widgets

A widget is a small gadget or control of your jQuery mobile application. Widgets can be very handy as they allow you to put your favorite applications on your home screen in order to quickly access them.

Type of Widgets

1 Buttons -It specifies clickable button that includes content like text or images.

- 2 Checkbox** - Checkboxes are used when more than one option is required to be selected.
- 3 Radiobox** - Radio buttons are used when out of many options, just one option is required to be selected.
- 4 Datepicker** - It is focused on the input to open an interactive calendar in a small overlay.
- 5 Collapsible** - Collapsible allows you to expand or collapse the content whenever clicked on it. It is very helpful for mobile device, which presents a brief content.
- 6 Controlgroup** - Controlgroups provide a set of buttons to specify a single block that looks like a navigation component.
- 7 Filterable** - By using the data-filter = "true" attribute, you can filter the children of any element.
- 8 Flipswitch** - Flip Switch allows you to turn off/on or true/false the switch by clicking on it for boolean style input.
- 9 Listview** - The purpose of listview component is to render complex and customized content in lists.
- 10 Loader** - The jQuery Mobile provides different ways of loading states to an element.
- 11 Navbar** - The navbar widget is a set of buttons which links you to other web pages or sections.
- 12 Panels** -Panels are used to display the DOM components in the box.
- 13 Popups** -Popup is a user interface that appears within a small window to display text, images, and other content.
- 14 Rangeslider** -Rangeslider widget provides you with a pair of handles allowing you to select a numeric value range.
- 15 Selectmenu** -A select menu provides various options in the form of dropdown list, from where a user can select one or more options.
- 16 Slider** -Slider allows you to choose a value by sliding the handle of the slider.
- 17 Table** -jQuery Mobile uses the table to represent the data in terms of rows and columns, i.e. displays the data in a tabular format.
- 18 Tabs** -The tabs widget is jQuery ui tabs widget's extension, which accepts all the methods and options.

19 Textinput -The <input> tag is used to declare an input element, a control that allows the user to input data.

20 Toolbar -The jQuery mobile toolbar widget allows you to create headers and footers.

Example

```
<div data-role="collapsible">
  <h3>Click Me</h3>
  <p>Hidden content</p>
</div>
```

jQuery Mobile - Events

jQuery Mobile allows to create dynamic web pages. By using events, you can set up event-driven process on the elements, which are triggered by the user's interaction such as mouse click, mouse hover on an element, key press on the keyboard, etc.

Following are some of the events for the mobile devices, which are supported by jQuery Mobile.

1 jQuery Mobile Events -It responds to user interaction when the user clicks on a certain page or hovers the mouse over an element, etc.

2 jQuery Touch Events -It provides touch events when the user touches the screen.

3 jQuery Scroll Events -It fires the scroll events when the user scrolls up and down.

4 jQuery Orientation Event -It triggers the orientation event when the user rotates the device vertically or horizontally.

5 jQuery Page Events -It provides the page events when the user hides, creates, loads, or unloads the pages.

Example

```
$(document).on("tap", function() {
  alert("Screen tapped!");
});
```

jQuery Mobile - Forms

Creation of forms is easy and very flexible, which are built with a combination of standardized form elements and buttons.

Type of Forms

1 Form Basic -jQuery Mobile provides powerful, easy, and versatile layout system for Forms, which combines form styles, input button, and slider support.

2 Form Inputs - The <input> tag is a control that allows the user to input data.

3 Form Select - In form of option, a dropdown list is provided for select menu.

4 Form Sliders - Slider allows you to choose a value by sliding the handle of the slider.

5 Refreshing and auto initialization of form elements - Refresh method is used to update the new state of form control by itself and updates the form control with JavaScript.

Example

```
<form>
  <label>Name:</label>
  <input type="text" name="name">
  <label>Gender:</label>
  <select>
    <option>Male</option>
    <option>Female</option>
  </select>
</form>
```

jQuery Mobile - Themes

It sets different types of theme on the buttons, navbars, blocks, links and so on. You can set the theme using *data-theme* attribute.

Following are use of the theme functionality in different areas which is supported by jQuery Mobile.

- 1 Themes** - It provides two different types of themes such as theme "a" and theme "b" to customize the look of the application.
- 2 Theming Header and Footer in Dialogs** - Sets the theme for header and footer in the dialog box.
- 3 Theming Buttons, Icons and Popups** - Specifies the theme for buttons, icons and popups.
- 4 Theming Buttons in Header and Footer** - Displays the theme for buttons in the header and footer.
- 5 Theming Navigation Bars** - Applies the theme for navigation bars in header or footer.
- 6 Theming Panels** - You can apply the theme for panel.
- 7 Theming Collapsible Button and Split Buttons** - Displays the theme for collapsible and split buttons.
- 8 Theming Lists and Collapsible Lists** - Displays the theme for lists and collapsible lists.
- 9 Collapsible Forms** - You can apply the theme for forms.

Example

```
<div data-theme="b">Styled Content</div>
```

jQuery Mobile - CSS Classes

jQuery CSS Classes

You can use different types of CSS classes to style the elements as described in the below sections.

```
<button class="ui-btn">Click</button>
```

Global Classes

The following classes can be used as global classes on jQuery Mobile widgets –

- 1 ui-corner-all** -It displays the elements with rounded corners.
- 2 ui-shadow** - It displays the shadow for the elements.
- 3 ui-overlay-shadow** - It displays the overlay shadow for the elements.
- 4 ui-mini** - It displays the smaller elements.

Button Classes

The following are lists button classes that are used with anchor or button elements –

- 1 ui-btn** - It specifies that the element will be styled as button.
- 2 ui-btn-inline** - It shows the button as inline element which saves the space as needed for the label.
- 3 ui-btn-icon-top** - It places the icon above the text.
- 4 ui-btn-icon-right** - It places the icon right of the text.
- 5 ui-btn-icon-bottom** - It places the icon below the text.
- 6 ui-btn-icon-left** - It places the icon left of the text.
- 7 ui-btn-icon-notext** - It shows the only icon.
- 8 ui-btn-a|b** - It displays the color of the button ("a" will be the default background color i.e. gray and "b" will change the background color to black).

Icon Classes

The following are lists of icon classes that are used with anchor or button elements –

- 1 ui-icon-action** - It shows the action icon.
- 2 ui-icon-alert** - It display the exclamation mark inside a triangle.
- 3 ui-icon-arrow-d-l** - It specifies down with left arrow.
- 4 ui-icon-arrow-d-r** - It specifies down with right arrow.
- 5 ui-icon-arrow-u-l** - It specifies up with left arrow.
- 6 ui-icon-arrow-u-r** - It specifies up with right arrow.
- 7 ui-icon-arrow-l** - It specifies the left arrow.
- 8 ui-icon-arrow-r** - It specifies the right arrow.
- 9 ui-icon-arrow-u** - It specifies the up arrow.
- 10 ui-icon-arrow-d** - It specifies the down arrow.
- 11 ui-icon-bars** - It shows the 3 horizontal bars one above the other.
- 12 ui-icon-bullets** - It shows the 3 horizontal bullets one above the other.
- 13 ui-icon-carat-d** - It displays the carat to down.

- 14 ui-icon-carat-l** -It displays the carat to left.
- 15 ui-icon-carat-r** -It displays the carat to right.
- 16 ui-icon-carat-u** - It displays the carat to up.
- 17 ui-icon-check** - It shows the checkmark icon.
- 18 ui-icon-comment** - It specifies the comment or message.
- 19 ui-icon-forbidden** - It displays the forbidden icon.
- 20 ui-icon-forward** - It specifies the forward icon.
- 21 ui-icon-navigation** - It specifies the navigation icon.
- 22 ui-icon-recycle** - It displays the recycle icon.
- 23 ui-icon-refresh** - It shows the refresh icon.
- 24 ui-icon-tag** -It indicates the tag icon.
- 25 ui-icon-video** - It indicates the video or camera icon.

Theme Classes

It provides two different types of themes such as theme "a" and theme "b" to customize the look of the application.

You can create your own theme classes by appending swatch letter (a-z). Following are theme classes which are specified from letter a to z.

- 1 ui-bar-(a-z)** - It displays the color for bar including headers, footers, and other bars in the page.
- 2 ui-body-(a-z)** - It displays the color for content block including listview, popups, sliders, panels, loaders, etc.
- 3 ui-btn-(a-z)** - It displays the color for button.
- 4 ui-group-theme-(a-z)** - It displays the color for controlgroups, listviews, and collapsible sets.
- 5 ui-overlay-(a-z)** - It displays the background color for popup, dialog, and page containers.
- 6 ui-page-theme-(a-z)** - It displays the color for pages.

Grid Classes

Following are grid classes that are used with equal width, no border, background, margin or padding.

1 ui-grid-solo 1 100%

2 ui-grid-a 2 50%/50%

3 ui-grid-b 3 33%/33%/33%

4 ui-grid-c 4 25%/25%/25%/25%

5 ui-grid-d 5 20%/20%/20%/20%/20%

jQuery Mobile - Data Attributes

Button

It specifies a clickable button that includes content like text or images using the class *ui-btn*. It is deprecated in version 1.4. Use the *ui-btn* attribute instead of using *data-role = "button"* attribute.

Following are the button elements used with data attribute.

1 data-corners -It defines whether the button should contain rounded corners or not.

2 data-icon -It defines the icon of the button.

3 data-iconpos -It defines the position of the icon.

4 data-iconshadow -It defines whether the icon of the button should contain shadow or not.

5 data-inline -It defines whether the button should be inline or not.

6 data-mini -It defines whether the button should display in smaller size or regular size.

7 data-shadow - It defines whether the button should contain shadow or not.

8 data-theme -It displays the theme color for the button.

Checkbox

Following are the checkbox elements used with *type = "checkbox"*.

1 data-mini - It defines whether the checkbox should display in smaller size or regular size.

2 data-role - It stops styling of checkboxes as buttons.

3 data-theme - It displays the theme color for the checkbox.

Collapsible

Following are collapsible elements used with ***data-role = "collapsible"*** attribute.

1 data-collapsed - It indicates whether the content should be closed or expanded.

2 data-collapsed-cue-text -It displays feedback for users with screen reader software.

3 data-collapsed-icon -It defines the icon of collapsible button.

4 data-content-theme -It displays the theme color for the collapsible content.

5 data-expanded-cue-text -It displays feedback for users with screen reader software.

6 data-expanded-icon -It displays the collapsible button when you expand the content.

7 data-iconpos -It defines the position of the icon.

8 data-inset -It defines whether the collapsible button should display with rounded corners and margin or not.

9 data-mini -It defines whether the collapsible buttons should display in smaller size or regular size.

10 data-theme -It displays the theme color for the collapsible button.

Collapsible Set

Following are the collapsible set elements used with the ***data-role = "collapsibleset"*** attribute.

1 data-collapsed-icon -It defines the icon of collapsible button.

2 data-content-theme -It displays the theme color for the collapsible content.

3 data-expanded-icon -It displays the collapsible button when you expand the content.

4 data-iconpos -It defines the position of the icon.

5 data-inset - It defines whether the collapsible button should display with rounded corners and margin or not.

6 data-mini -It defines whether the collapsible buttons should display in smaller size or regular size.

7 data-theme -It displays the theme color for the collapsible button.

Controlgroup

Following are Controlgroup elements used with ***data-role = "controlgroup"*** attribute

1 data-exclude-invisible - It defines whether to exclude invisible children in the assignment of rounded corners.

2 data-mini - It defines whether the group should display in smaller size or regular size.

3 data-theme - It displays the theme color for the controlgroup.

4 data-type - It indicates whether the group should display in horizontal or vertical format.

Dialog

Following are dialog elements used with ***data-dialog="true"*** attribute.

1 data-close-btn -It defines the position of the close button.

2 data-close-btn-text - It defines the text for the close button.

3 data-corners - It defines whether dialog should display with rounded corners or not.

4 data-dom-cache -It indicates whether DOM cache must clear or not for individual pages.

5 data-overlay-theme -It defines the overlay color of the dialog page.

6 data-theme -It defines the theme color of the dialog page.

7 data-title -It defines the title of the dialog page.

Enhancement

Following are lists of enhancement elements used with *data-enhance="false"* or *data-ajax = "false"* attribute.

1 data-enhance -You can style the page by setting this attribute to "true". You cannot style the page if it is set to "false".

2 data-ajax - It indicates whether pages must load from Ajax or not.

Fixed Toolbar

Following are the toolbar elements used with *data-position = "fixed"* attribute.

1 data-disable-page-zoom -It defines whether the user is able to scale/zoom the page or not.

2 data-fullscreen -It defines toolbars must be positioned at the top and/or bottom.

3 data-tap-toggle -It indicates whether the user can toggle toolbar-visibility on taps or not.

4 data-transition -It shows a transition effect when you tap or click the element.

5 data-update-page-padding -It updates the padding of page by using resize, transition and update layout events.

6 data-visible-on-page-show -It defines toolbar-visibility when the parent page is shown.

Flip Toggle Switch

Following are flip toggle elements used with *data-role = "flipswitch"* attribute

1 data-mini - It defines whether the switch should display in smaller size or regular size.

2 data-on-text - It defines the "on" text on the flip switch.

3 data-off-text - It defines the "off" text on the flip switch.

Footer

The following are footer elements used with *data-role="footer"* attribute –

1 data-id -It defines the unique ID.

2 data-position - It defines whether the footer should be positioned at the bottom or inline with page content.

3 data-fullscreen -It defines whether the footer should be positioned at the bottom and over the page content or not.

4 data-theme - It defines the theme color of the footer.

Header

Following are the lists of the header elements used with *data-role = "header"* attribute.

1 data-id -It defines the unique ID.

2 data-position - It defines whether the header should be positioned at the bottom or inline with the page content.

3 data-fullscreen -It defines whether the header should be positioned at the bottom and over the page content or not.

4 data-theme -It defines the theme color of the header.

Inputs

Following are lists of the input elements used with *type = "text|search|etc"* attribute.

1 data-clear-btn -It defines whether the input element should contain clear button or not.

2 data-clear-btn-text - It defines the text for the clear button.

3 data-mini -It defines whether input should display in smaller size or regular size.

4 data-role -It stops styling input or text areas as buttons.

5 data-theme -It defines the theme color of the input element.

Link

Following are the lists of the link elements used with jQuery Mobile.

1 data-ajax -It indicates whether the pages must be loaded through Ajax or not

2 data-direction -It is used for reverse transition. reverse

3 data-dom-cache -It indicates whether jQuery DOM cache must be clear or not for pages.

4 data-prefetch -It is used to prefetch the pages into DOM.

5 data-rel -It specifies the behavior of the link.

6 data-transition -It defines the transition from one page to another.

7 data-position-to - It defines the position of the popup boxes.

List

Following is list of elements used with ***data-role = "listview"*** attribute.

1 data-autodividers - It divides the list automatically.

2 data-count-theme - It defines the theme color of the count element.

3 data-divider-theme - It defines the theme color for list divider.

4 data-filter - It is used to filter the list values in the search box.

5 data-filter-placeholder - It defines some text inside the search box.

6 data-filter-theme - It defines the theme color for search filter.

7 data-icon - It provides the icon for the list.

8 data-inset -It defines whether list should display with rounded corners and margin or not.

9 data-split-icon -It defines the icon for split button.

10 data-split-theme - It defines the theme color for split button.

11 data-theme -It defines the theme color for the list.

List Item

Following is list of item elements used with ***data-role = "listview"*** attribute.

1 data-filtertext - It is used to filter the list values using the text in the search box.

2 data-icon - It provides the icon for the list item.

3 data-role - It defines the divider for list items.

4 data-theme - It defines the theme color for the list item.

Navbar

Following is lists of navbar elements used with *data-role = "navbar"* attribute.

1 data-icon - It provides the icon for the list item.

2 data-iconpos - It defines the position for the icon.

Page

Following is lists of page elements used with *data-role = "page"* attribute.

1 data-dom-cache - It indicates whether DOM cache must clear or not for individual pages.

2 data-overlay-theme - It defines the overlay color of the dialog pages.

3 data-theme - It defines theme color for the page.

4 data-title - It provides the title for the page.

5 data-url - It is used to updating the URL.

Popup

Following is lists of the popup elements used with *data-role = "popup"* attribute.

1 data-corners - It defines whether the popup should display with rounded corners and margin or not.

2 data-dismissible -It defines the whether popup should be close by clicking outside or not.

3 data-history - It defines the whether popup should display the history of item when opened.

4 data-overlay-theme - It defines the overlay color of the popup box.

5 data-shadow - It displays the shadow for the popup box.

6 data-theme - It defines theme color for the popup box.

7 data-tolerance - It defines the edges of the window.

Radio Button

Following are the lists of the radio button elements used with *type = "radio"* attribute.

- 1 **data-mini** - It defines whether the button should display in smaller size or regular size.
- 2 **data-role** - It stops the styling of radio buttons as enhanced buttons.
- 3 **data-theme** - It defines the theme color for the radio button.

Select

Following is the lists of select elements used with jQuery Mobile.

- 1 **data-icon** -It provides the icon for the select element.
- 2 **data-iconpos** -It defines the position of the icon.
- 3 **data-inline** -It defines whether the button should be inline or not.
- 4 **data-mini** -It defines whether select should display in smaller size or regular size.
- 5 **data-native-menu** - It use custom menu when it has been set to false.
- 6 **data-overlay-theme** - It defines the overlay color for the custom select menu.
- 7 **data-placeholder** - It is used to set an option element of non-native select.
- 8 **data-role** - It stops the styling of select elements as buttons.
- 9 **data-theme** -It displays the theme color for the select elements.

Slider

Following are the slider elements used with *type = "range"* attribute.

- 1 **data-highlight** -It highlights the slider.
- 2 **data-mini** -It defines whether the slider should display in smaller size or regular size.
- 3 **data-role** -It stops the styling of slider controls as buttons.
- 4 **data-theme** -It displays the theme color for the slider control.
- 5 **data-track-theme** -It displays the theme color for the slider track.

Advantages of jQuery Mobile

- It is easy to learn and develop applications if you have knowledge of HTML5, CSS3 features.
- It is cross-platform and cross-browser compatible so you don't have to worry about writing different code for each device resolution.
- You can create the custom theme using ThemeRoller without writing the line of code. It supports all HTML5 browsers.
- It uses HTML5 along with JavaScript for easy development of web applications.
- It is built in a way that allows the same code to automatically scale from the mobile screen to desktop screen.

Disadvantages of jQuery Mobile

- There are limited options for CSS themes, so sites can look similar which are built by these themes.
- Applications which are developed using jQuery Mobile are slower on mobiles.
- It becomes more time consuming when you combine jQuery mobile with other mobile frameworks.
- Difficult to provide complete customized visual design.
- All the features in a device cannot be accessed by JavaScript in a browser.